

Personal Statement

I am currently pursuing a Master's in Computer Engineering at the College of Electrical and Mechanical Engineering (CoEME), NUST. Alongside my studies, I am also serving as a Research Assistant at the BIOMISA Laboratory, where I am working on a cutting-edge research project focused on advancing semantic segmentation for baggage threat detection. This project leverages an incremental Task-Agnostic Meta-Learning (iTAML) approach to develop a robust model capable of accurately identifying and segmenting threats (such as guns and knives) within X-ray images. Unlike conventional deep learning methods that require extensive retraining for each new object, the iTAML-based framework enables rapid adaptation to novel threats with only a few examples. I hold a BS degree in Electrical and Electronics Engineering, where my award-winning Final Year Project, titled "Improved Intelligent Surveillance System," applied CNN-LSTM techniques for violence detection. Additionally, I have authored two out of three IEEE conference papers as the lead author and a review paper reflecting my commitment to advancing innovative research in the field.

Education

Master of Science in Computer Engineering

Sep 2024 – June 2026

College of Electrical and Mechanical Engineering, NUST

CGPA 3.61

Thesis Title Semantic Segmentation using Task Agnostic Meta Learning for Baggage Threat Detection

Major Subjects Deep Learning, Digital Image Processing, Adv Programming in Python, Machine Learning for Embedded Systems, Adv Computer Architecture, Adv Digital Signal Processing, Adv Digital System Design

Bachelor of Science in Electrical (Electronics) Engineering

Sep 2018 – Aug 2022

Comsats University Islamabad, Abbottabad Campus

CGPA 3.06/4 (76.5%)

Thesis Title Improved Intelligent Surveillance System

Major Subjects Introduction to Computer Programming, Object Oriented Programming, Control Systems, Digital Signal Processing, Digital System Design, VLSI Design, Digital Logic Design.

Experience

1. RISETech (June 2024 - August 2024)

As an intern, my role was to develop machine learning techniques for detecting Diabetic Retinopathy and Glaucoma. My responsibilities included code implementation, model training, and performance testing.

2. Teacher Assistance (March 2025 – Dec 2025)

As a part time teacher assistant my main role was to prepare bilingual (Chinese and English) lecture slides for BS Students.

- Subjects include Electrical Safety, Power System Relaying, Object Oriented Programming

3. BIOMISA (Sep 2024 – till now)

As a research assistant, my role involved developing a novel meta-learning technique for task-incremental learning. My responsibilities included:

- Conducting an extensive literature survey on incremental meta-learning.

- Implementing task-incremental meta-learning for classification tasks.
- Integrating SegFormer as a backbone for segmentation task.
- Training and evaluating models on different baggage datasets.

Publications

Review Journal

1. **W. Iqar**, K. Khurshid, S. Khan and N. Saeed, "The AIoT Ecosystem for Next-Generation Satellite Systems," *Physical Communication*, 2025. [IF=2.2]

Conferences

1. **W. Iqar**, A. Shahzad, W. Hameed and M. Z. Abidien, "A Real-time Sequence Based Human Activity Detection System," 2023 International Multi-disciplinary Conference in Emerging Research Trends (IMCERT), Karachi, Pakistan, 2023, pp. 1-5, doi: 10.1109/IMCERT57083.2023.10075257. <https://ieeexplore.ieee.org/document/10075257>
2. **W. Iqar**, M. Z. Abidien, W. Hameed and A. Shahzad, "CNN-LSTM Based Smart Real-time Video Surveillance System," 2022 14th International Conference on Mathematics, Actuarial Science, Computer Science and Statistics (MACS), Karachi, Pakistan, 2022, pp. 1-5, doi: 10.1109/MACS56771.2022.10023018. <https://ieeexplore.ieee.org/abstract/document/10023018>
3. R. Iqar, **W. Iqar** and S. Mazhar, "Design Validation and Regulation of Boost Converter: A Bond graph Approach," 2023 3rd International Conference on Digital Futures and Transformative Technologies (ICoDT2), Islamabad, Pakistan, 2023, pp. 1-6, doi: 10.1109/ICoDT259378.2023.10325813. <https://ieeexplore.ieee.org/document/10325813>

Coursera Certificates

1. Fundamentals of Reinforcement Learning

An online course authorized by University of Alberta, Alberta Machine Intelligence Institute

Link: <https://coursera.org/share/915ad25aba43e480677c01075bc962e8>

2. AI for Everyone

This course focuses on the objectives of neural networks, machine learning, and deep learning. Its objectives include determining how to detect AI potential within your own organization.

Link: <https://coursera.org/share/852416d39d659fec3935211629f9fb72>

Awards & Certificates

1. Certificate of Appreciation as Paper Presenter at IMCERT 2023 Conference
2. IGNITE funding Award from Federal Ministry of Information Technology. (NGIRI-2022-12025)
3. Registered Engineer from Pakistan Engineering Council (Reg No: ELECT/99345)

Language Skills

1. English
IELTS Score Overall 6.5 (recently expired)
2. Urdu Native

Projects

1. Intelligence Surveillance System.

A CNN-LSTM approach can detect suspicious activities in real time. A model is trained on this technique and is implemented on Raspberry Pi for Real-Time application. This project works as a standalone system with 10-13 FPS in real-time which acts as a surveillance system capable of suspicious activities.

2. Weapon Detection

YOLOv4 technique is utilized and trained on custom dataset to detect weapons for real-time applications. This

model is also capable of differentiating between real weapons and toy weapons.

3. Baggage Segmentation

A segmentation model (SegFormer) has been trained and deployed on the Jetson Nano for real-time security applications. The deployed model is capable of classifying images within approximately three seconds, enabling efficient and reliable performance in practical scenarios.

4. Modeling and implementation of 8-bit CPU.

Complete modeling and architecture of 8-bit CPU is done in *Verilog*. This CPU is capable of performing 14 analytical operations.

5. Multiplayer Ludo Game using Java.

An object-oriented programming game development using Java as programming language.

Digital Skills

Python | Google COLAB | Java | C++ | MATLAB | ModelSim | Verilog | Xilinx | FPGA | Multisim | Microsoft Visio | 3D Blender

Workshops & Seminars

1. 2nd International Workshop on Recent Trends in Electric Machines.
2. Artificial Intelligence: Implication for technology & Business Strategy.
3. Cellular Network (2G to 5G Technology) Future Applications.
4. Solar and Wind Power Potential in Pakistan.

Recommendations

- **Dr. Muhammad Usman Akram** (Professor & Director BIOMISA Research Group)
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